

Code No: 114CQ**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech II Year II Semester Examinations, February - 2024****DATABASE MANAGEMENT SYSTEMS****(Computer Science and Engineering)****Time: 3 hours****Max. Marks: 75****Note:** i) Question paper consists of Part A, Part B.

ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.

iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A**(25 Marks)**

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|------|--|-----|
| 1.a) | What is a database? | [2] |
| b) | Quote example for aggregation in ER model. | [3] |
| c) | What is an unsafe query? | [2] |
| d) | What is a trigger? | [3] |
| e) | Define functional dependency. | [2] |
| f) | Give example for spurious tuples. | [3] |
| g) | Define schedule. | [2] |
| h) | What are the benefits of multiple granularity? | [3] |
| i) | Define index. | [2] |
| j) | What is static hashing? | [3] |

PART - B**(50 Marks)**

2. How does database management system provide data abstraction? Explain in detail.[10]
OR
3. What is an integrity constraint? Discuss various types of integrity constraints for DBMS. [10]
4. Illustrate all primitive operators available in Relational Algebra for data accessing. [10]
OR
5. Consider the following database schema to write queries in SQL:
Employee(Eid, Ename, Ecity)
Company(Cno, Cname, Ccity)
Works_for(Eid, Cno, Salary)
a) Find the employees hailing from Kolkata.
b) Find the employees working for Wipro.
c) Find the average salary of employees working for IBM. [3+3+4]
6. Explain the need of schema refinement in database design with suitable examples. [10]
OR
7. Make a comparison of Boyce Codd normal form and Fourth normal form. [10]

8. Discuss the concept of serializability with suitable schedules. [10]

OR

9. Demonstrate the working of validation based protocol. [10]

10. Compare tree based indexing with hash based indexing. [10]

OR

11.a) Illustrate dense indexing.

b) Explain insertion operation on hash file. [5+5]

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